

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – SHORT ANSWER TEST

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 – Short Answer Test
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	K. Ragani	Reg. No.:	192524364
Section / Batch:	B.Tech - AI & DS	Date:	02/09/2026

Code of Conduct

I, K. Ragani (192524364) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Ragani

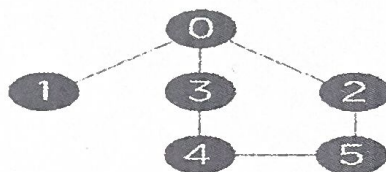
Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Annexure A - Short Answer Test

- Perform the following operations in the Singly Linked List for the following input.
Input: 22-> 77-> 55-> 88->
 - Insert a new-node 110 at the end of the list.
 - Insert a new-node 101 at the position 4 of the list.
 - Delete a node at the beginning of the single linked list
 - Delete a node at the position 3 of the single linked list.
- Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10 (), 4) Push (90), 5) Push (36), 6) Push (11), 7) Push (88), 8) Pop (), 9) Pop (). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
- Convert the following infix into a postfix expression. $\rightarrow B - (C/D + (E \% F * G) / H) * J$
- Evaluate the given postfix expression. $53+62/*35*+$
- Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
- Construct a 2-3-4 Tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.
- Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42 and 19. At each stage of heap tree construction, ensure heap properties are satisfied.
- Perform the operations onto splay tree (draw after each operation): insert (3), insert (9), insert (12), insert (55), insert (1), insert (3), delete (3), delete (9), delete (60), insert (3), search (12).
- Describe the algorithm to sort the elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix sort. Explain with each phase?
- Consider the following graph & construct BFS.



Graph with six nodes

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with minor gaps	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with minor errors	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with minor gaps	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & Completion	10	Completes all questions accurately within the time	Completes most questions on time	Requires additional time	Unable to complete the test

S Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Perform the following operations in the singly linked list for the following input.

Input : 22 → 77 → 55 → 88 →

- i) Insert a new node 110 at the end of the list

Traverse to the last node (88), set its next pointer to the new node 110.

22 → 77 → 55 → 88 → 110

- ii) Insert a new node 101 at the position 4 of the list.

Position 4 means the new node becomes the 4th node, pushing the current 4th node (88) to position 5. Traverse to position 3 (55), link it to 101, and link 101 to the old node at position 4 (88).

22 → 77 → 55 → 101 → 88 → 110

- iii) Delete a node at the beginning of the singly linked list.

Move head pointer to the second node, removing 22.

77 → 55 → 101 → 88 → 110

- iv) Delete a node at the position 3 of the singly linked list. Position 3 in the current list is 101. Link position 2 (55) directly to position 4 (88), bypassing 101.

77 → 55 → 88 → 110 ← final list

2. Rotations in each step of insertion. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10(), 4) Push (90), 5) Push (36), 6) Push (11), operations and identify where the top

Initial stack : Empty

1. Push (90)

TOP $\rightarrow$ 90

2. Push (36)

TOP $\rightarrow$ 36
90

3. Push (11)

TOP $\rightarrow$ 11
36
90

4. Push (88)

TOP $\rightarrow$ 88
11
36
90

5. POP()

88 is removed

TOP $\rightarrow$ 11
36
90

6. POP()

11 is removed

TOP $\rightarrow$ 36
90

Final stack

TOP $\rightarrow$ 36
90

TOP element = 36

Conclusion : stack follows LIFO (Last in, First out). The last inserted element is remove first.

3. convert the following infix into a postfix expression $\rightarrow B - (C/D + (E + F * G) / H)^j$

Expression : $B - (C/D + (E + F * G) / H) * J$

Token	Action	Operator stack	Output so far
B	output	(empty)	B
-	push	-	B
(	push	- (	B
C	output	- (	BC
/	push	- (/	BC
D	output	- (/	BCD
+	pop / , push +	- (+	BCD / E
(	push	- (+ (	BCD / E F + G
E	output	- (+ (	BCD / E F + G *
*	push	- (+ (*	BCD / E F + G * H
F	output	- (+ (*	BCD / E F + G * H /
*	pop *, push *	- (+ (*	BCD / E F + G * H / +
G	output	- (+ (*	BCD / E F + G * H / +
)	pop *, pop (	- (+	BCD / E F + G * H / + J
end	output	(empty)	BCD / E F + G * H / + J *

post fix result : $BCD / E F + G * H / + J *$

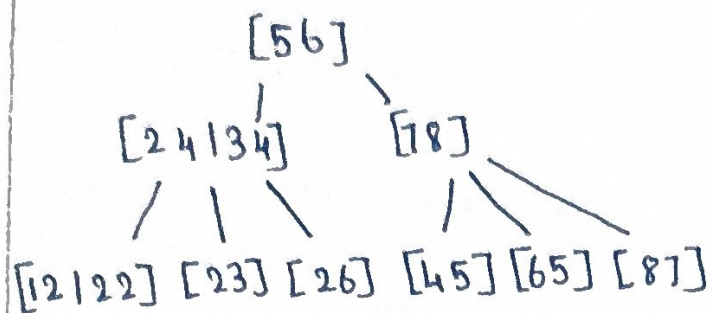
4. Evaluate the given postfix expression $53 + 62 / * 35 ^ +$
 Expression: $53 + 62 / * 35 ^ +$

Token	Action	Stack after action
5	push	5
3	push	5, 3
+	POP 3, 5 $\rightarrow 5+3=8$, Push	8
6	push	8, 6
2	push	8, 6, 2
/	POP 2, 6 $\rightarrow 6/2=3$, Push	8, 3
*	POP 3, 8 $\rightarrow 8*3=24$, Push	24
3	push	24, 3
5	push	24, 3, 5
*	POP 5, 3 $\rightarrow 3*5=15$, Push	24, 15
+	POP 15, 24 $\rightarrow 24+15=39$, Push	39

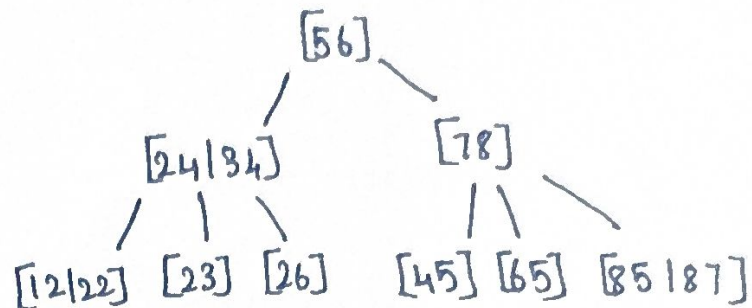
Result = 39

5. construct a B-tree of order 3 by inserting numbers 34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95.

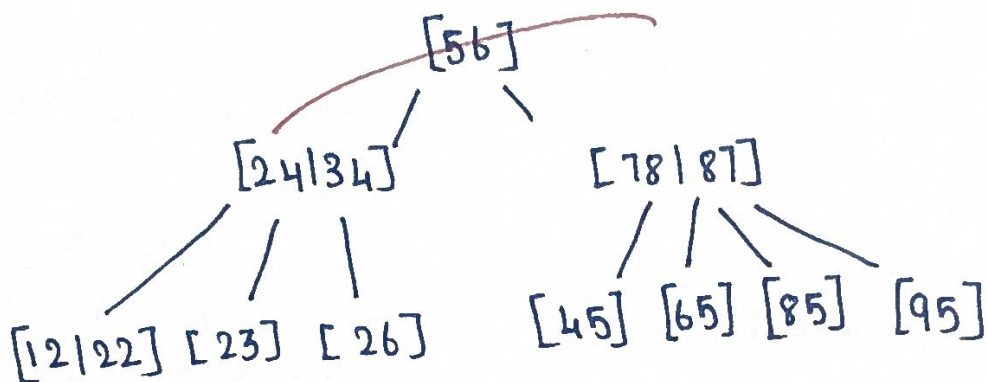
11. Insert 65



12. Insert 85



Final answer: Insert 95



6

construct a 2-3-4 tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.

1. Insert 45

[45]

2. Insert 36

[36|45]

3. Insert 76

[36|45|76]

4. Insert 23

[45]

[23|36] [76]

5. Insert 89

[45]

[23|36] [76|89]

6. Insert 115

[45|89]

[23|36] [76] [115]

7. Insert 98

[45|89]

[23|36] [76] [98|115]

8. Insert 39

[36|45|89]

[23] [39] [76] [98|115]

1. Insert 34

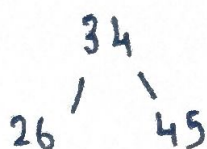
[34]

2. Insert 26

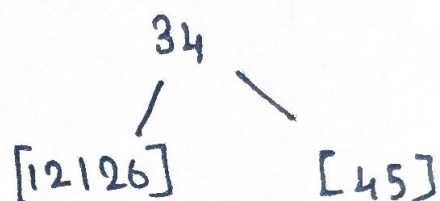
[26][34]

3. Insert 45

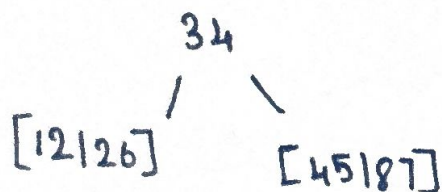
[26|34|45]



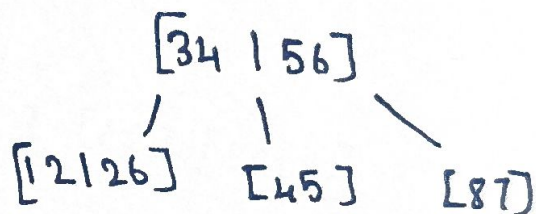
4. Insert 12



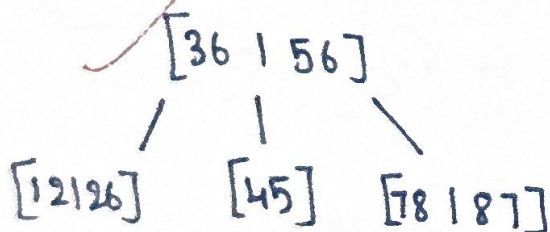
5. Insert 87



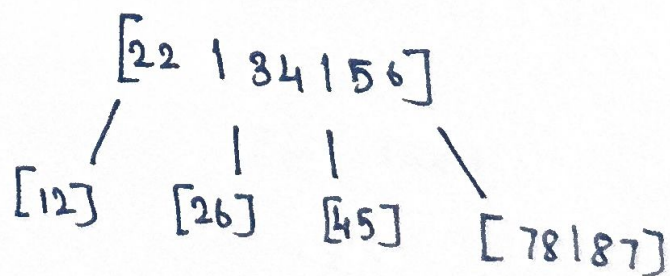
6. Insert 56



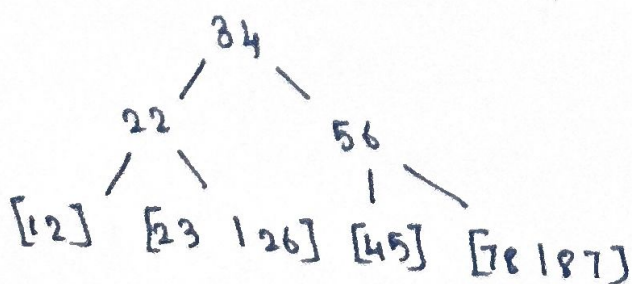
7. Insert 78



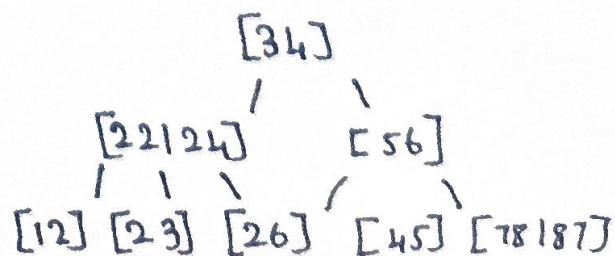
8. Insert 22



9. Insert 23

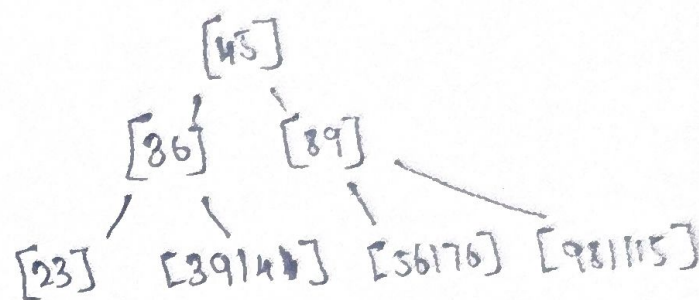
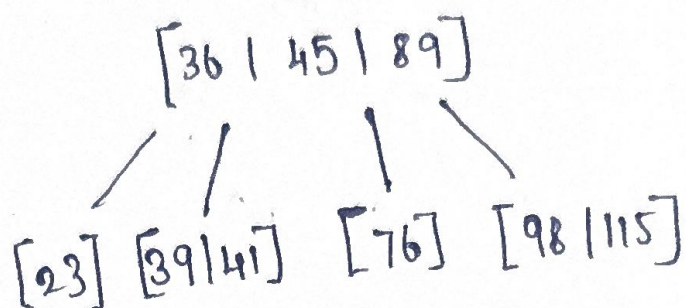


10. Insert 24

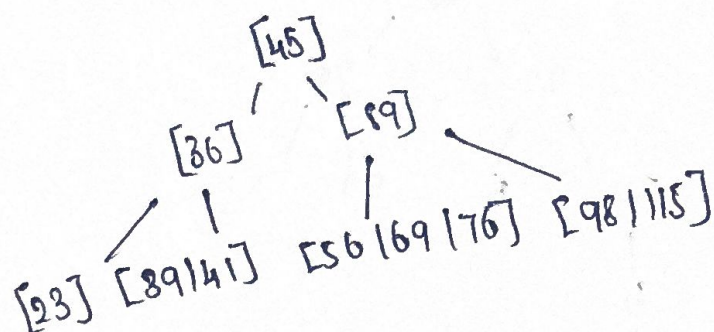


9. Insert 41 → Split left node

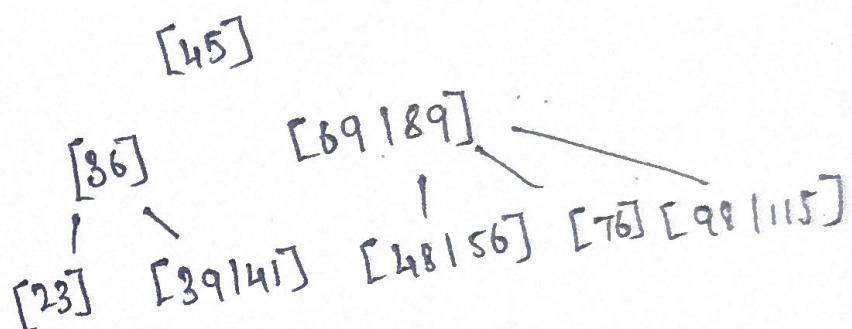
10. Insert 56 → roots splits



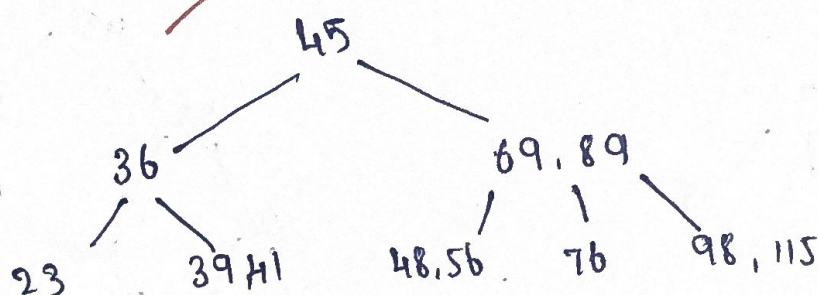
11. Insert 69



12. Insert 48. final tree



Ans:



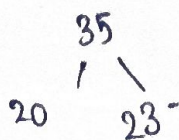
7. Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42 and 19.

1. Insert 20

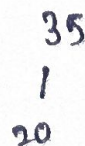
20

9

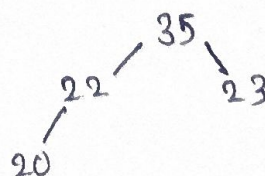
3. Insert 23



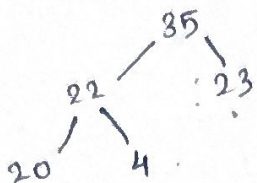
2. Insert 35



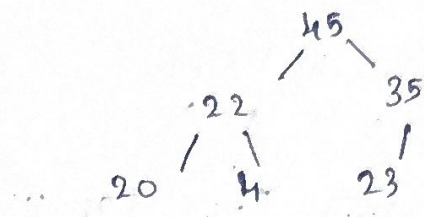
4. Insert 22



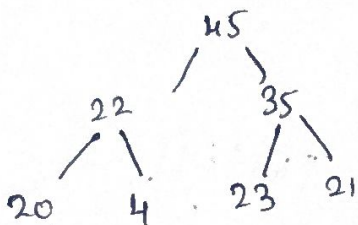
5. Insert 4



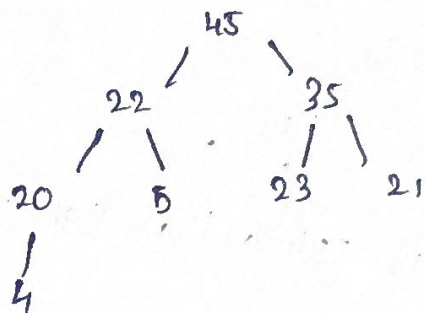
6. Insert 45 → 45 moves up



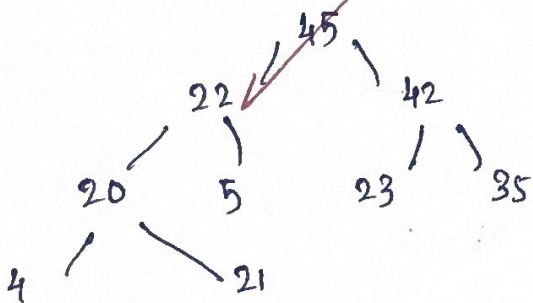
7. Insert 21



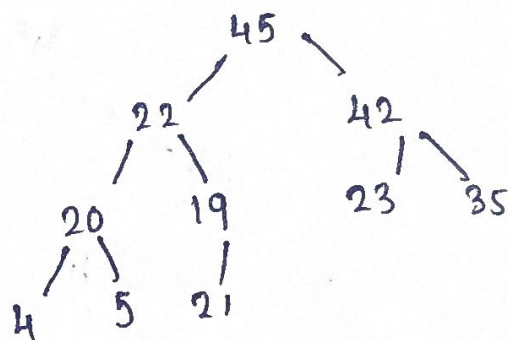
8. Insert 5



9. Insert 42 → 42 moves up



10. Insert 19



8. Perform the operation onto splay tree (draw after each operation):
 insert (3), insert (9), insert (12), insert (55), insert (1), insert (3)
 delete (9), delete (60), insert (3), search (12).

1. Insert (3)

3

2. Insert (9)

9

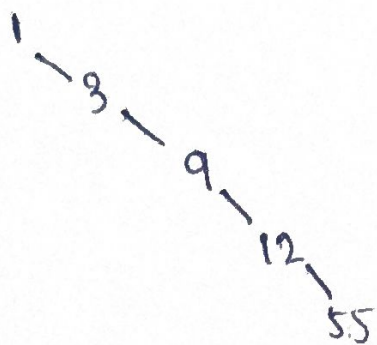
3. Insert (12)

3 / 9 / 12

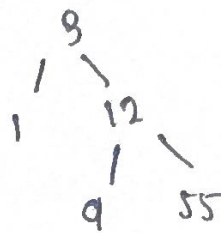
4. Insert (55)

3 / 9 / 12 / 55

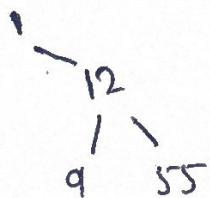
5. Insert (1)



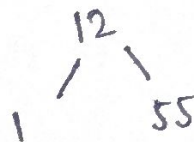
6. Insert



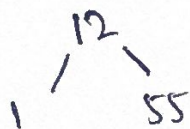
7. Delete (3)



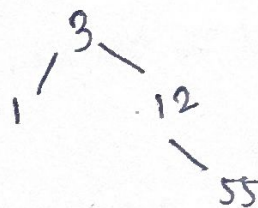
8. Delete (9)



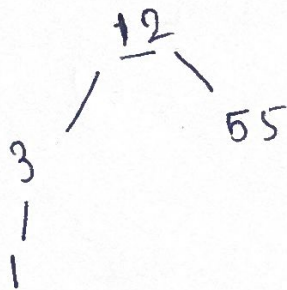
9. Delete (60)



10. Insert (3)



Final: search (12)



9 Describe the algorithm to sort the elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix sort. Explain with each phase?

Radix sort sorts numbers digit by digit, starting from the units digit.

Given :

170, 45, 75, 90, 802, 24, 2, 66

Phase 1 - unit digit

After sorting according to the units digit.

170, 90, 802, 2, 24, 45, 75, 66

Phase 2 - tens digit

After sorting according to the tens digit :

802, 2, 24, 45, 66, 170, 75, 90

Phase 3 - hundreds digit

After sorting according to the hundreds digit :

2, 24, 45, 66, 75, 90, 170, 802

Final sorting order

2, 24, 45, 66, 75, 90, 170, 802

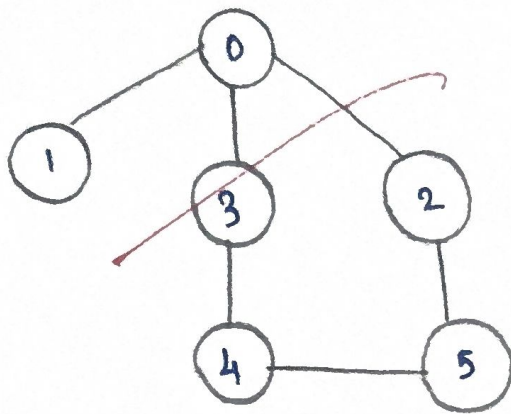
Algorithm

1. Find the maximum number of digits
2. Sort the numbers based on the units digit.
3. Sort again based on the tens digit.
4. Sort again based on the hundreds digit.
5. Continue until all digits are processed.

6. The numbers are now sorted

Result : The given numbers are sorted in ascending order using Radix sort.

10 consider the following graph & construct BFS.



Answer

BFS visits each vertex level by level using a queue.

starting vertex: 0

Traversal steps

1. start with 0
2. visit its adjacent vertices: 1, 3, 2
3. visit the next unvisited vertex 4.
4. visit the next unvisited vertex 5.
5. All vertices are now visited.

BFS Traversal

0 → 1 → 3 → 2 → 4 → 5

Result :

Therefore, the BFS traversal of the given graph is :

0, 1, 3, 2, 4, 5